



Encoders and Late Interaction: Open Source Search at LightOn



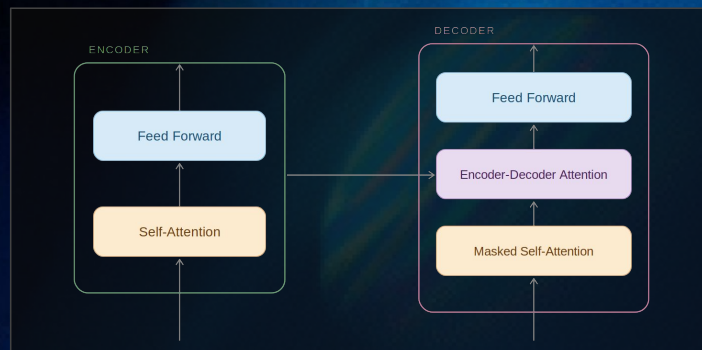
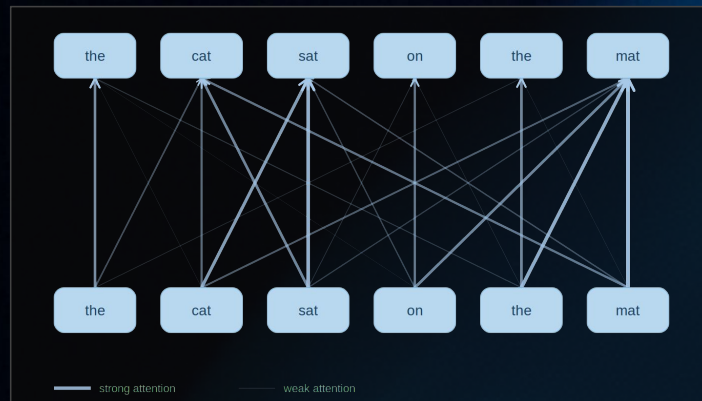
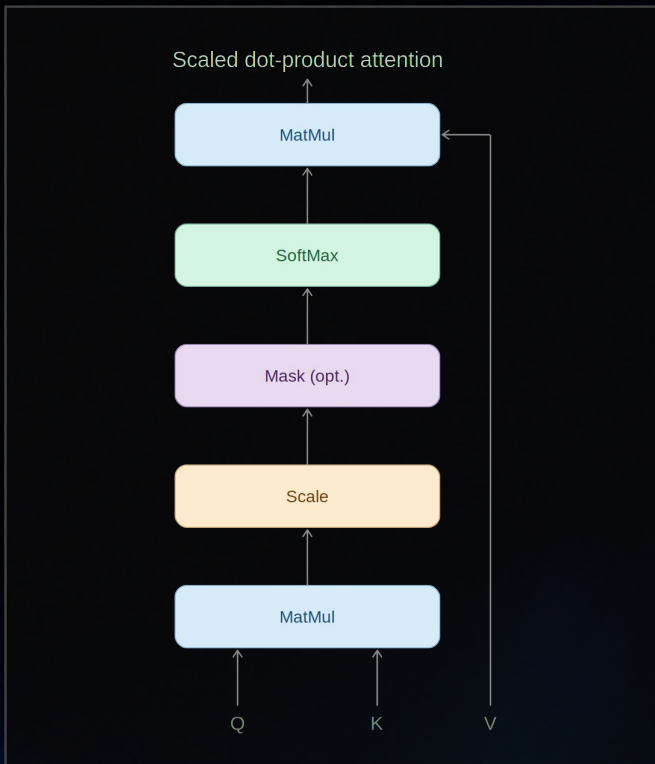
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Encoders

Attention is all you need

- Attention allows tokens to **attend** to others to get information from them → **information flow**
- Masked?



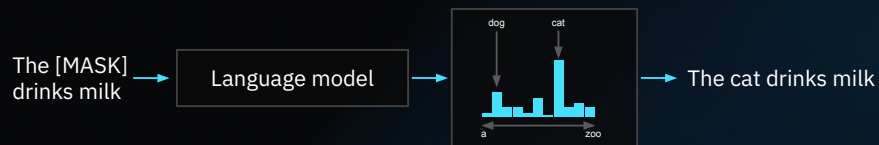
Encoder vs Decoder

- All tokens attend to all tokens
- Autoregressive generation: predict token $n+1$
 - But if n sees $n+1$...
- Mask future tokens



Decoder

- Predict [MASK] tokens using full context



Encoder

The rise of decoder-only

- Original transformers: **encoder-decoder** (T5)
- Generation-oriented: **decoder-only** (GPT)
- Discrimination-oriented: **encoder-only** (BERT)
- All tasks can be casted as generative tasks + generation is a good pre-training task
 - **All-in on decoders!**

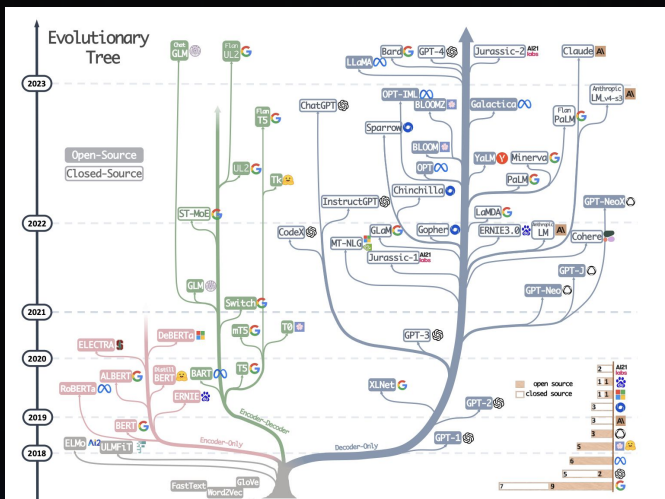


Fig. 1. The evolutionary tree of modern LLMs traces the development of language models in recent years and highlights some of the most well-known models. Models on the same branch have closer relationships. Transformer-based models are shown in non-grey colors: decoder-only models in the blue branch, encoder-only models in the pink branch, and encoder-decoder models in the green branch. The vertical position of the models on the timeline represents their release dates. Open-source models are represented by solid squares, while closed-source models are represented by hollow ones. The stacked bar plot in the bottom right corner shows the number of models from various companies and institutions.

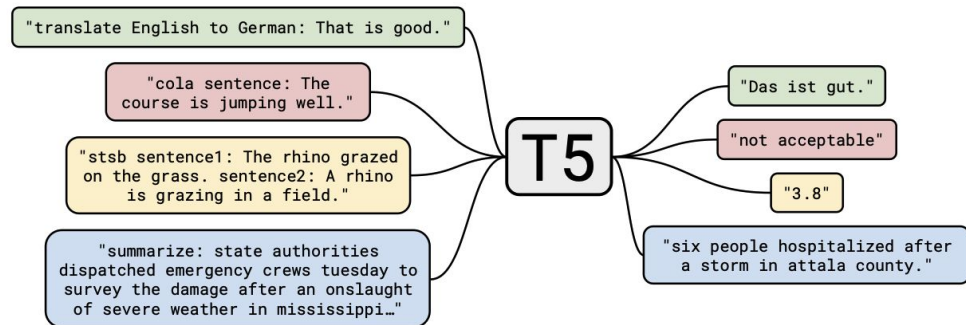


Figure 1: A diagram of our text-to-text framework. Every task we consider—including translation, question answering, and classification—is cast as feeding our model text as input and training it to generate some target text. This allows us to use the same model, loss function, hyperparameters, etc. across our diverse set of tasks. It also provides a standard testbed for the methods included in our empirical survey. “T5” refers to our model, which we dub the “Text-to-Text Transfer Transformer”.

And the fall of encoder-only?

- Large part of the most downloaded models on Hugging Face are encoders
 - Early 2025: over half of the 100 top hundred models
- Few new models, but still used a lot

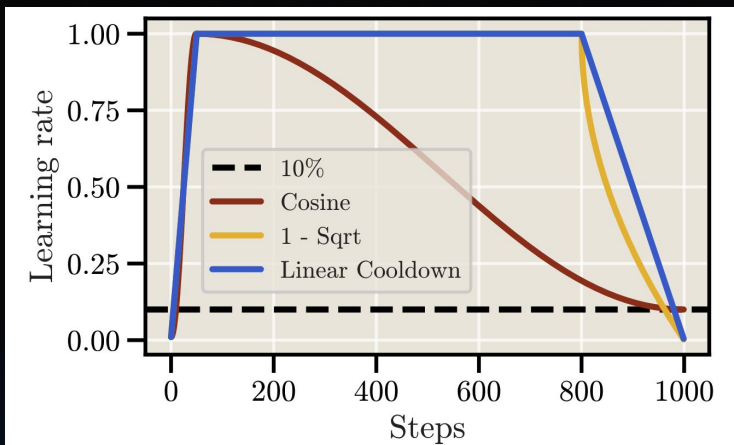
The screenshot shows the Hugging Face Models page with 2,695,949 models. The list is sorted by 'Most downloads'. The models are displayed in a grid with columns for model name, task, and download statistics. The models listed include:

Model Name	Task	Downloads
sentence-transformers/all-MiniLM-L6-v2	Sentence Similarity	22.7M
google/bert-base-uncased	Fill-Mask	67.1M
google/electra-base-discriminator	Image Classification	4.55k
Falconsai/nsfw_image_detection	Image Classification	1.01k
sentence-transformers/all-mpnet-base-v2	Sentence Similarity	27.5M
time/mobilenetv2_small_100_lamb_in1k	Image Classification	54
FacebookAI/roberta-large	Fill-Mask	23.3M
sentence-transformers/paraphrase-multilingual-MiniLM-L6-v2	Sentence Similarity	22.8M
Qwen/Qwen2.5-7B-Instruct	Text Generation	88
laion/clap-htsat-fused	Audio Classification	21.7M
FacebookAI/xlm-roberta-base	Fill-Mask	19.8M
openai/clip-vit-base-patch32	Zero-Shot Image Classification	19.1M
amazon/chronos-2	Time Series Forecasting	18.6M
Bingsu/adetailer	Image Classification	18M
BAAI/bge-m3	Sentence Similarity	17.8M
pyannote/wespeaker-voxceleb-resnet34-LM	Text Ranking	15M
pyannote/segmentation-3.0	Voice Activity Detection	14.8M
cross-encoder/ms-marco-MiniLM-L6-v2	Text Ranking	14.2M
Qwen/Qwen3-VL-2B-Instruct	Image-Text-to-Text	13.1M
pyannote/speaker-diarization-3.1	Automatic Speech Recognition	13.1M
colbert-ir/colbertv2.0	Text Generation	12.2M
Xenova/paraphrase-multilingual-MiniLM-L12-v2	Feature Extraction	12.2M
Qwen/Qwen3-0.6B	Text Generation	12M
omni-research/Tarsier2-Recap-7b	Text Generation	11.1M
openai-community/gpt2	Text Generation	10.1M
FacebookAI/roberta-base	Fill-Mask	10.1M

ModernBERT: BERT... but modern

Modern recipe

- Trained on **2T tokens** including web documents, code, and scientific literature
- **Modern tokenizer** supporting code
- **Infinite LR schedule** for easy continual pretraining (BioClinical ModernBERT)
 - **Context length expansion** to 8K (+ RoPe)

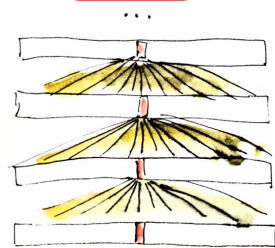


Modern architecture

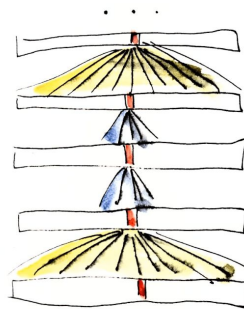
- **Global-local** attention alternation
- Full model **unpadding** (+ sequence packing)
- **Hardware aware** shapes
- Flash Attention 3

Attention patterns considering a single token

(shown in red)



Global attention on every layer: all tokens attend to all other tokens

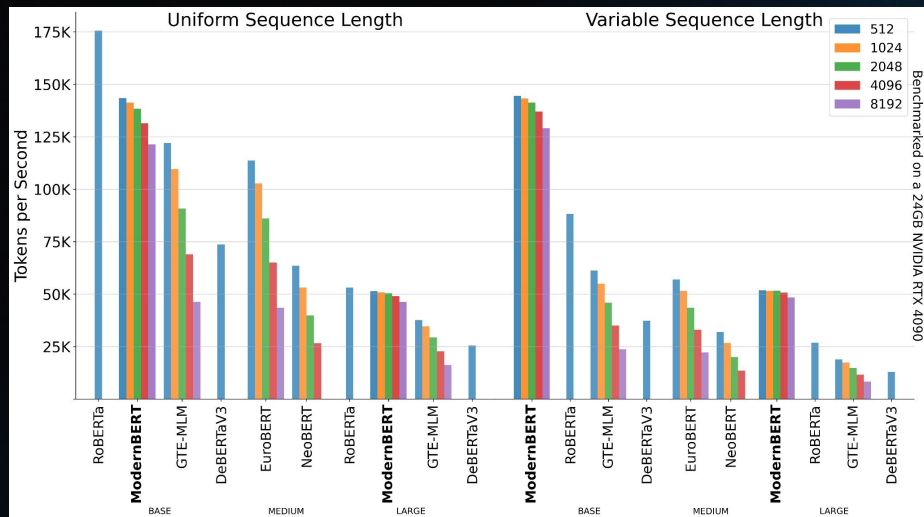


Alternating global and local attention. In local attention layers, a token only attends to those in a small window around it

Modern performance

- First masked language modeling only model to **beat DeBERTa on GLUE** (at a *much* lower cost/higher speed)
- State-of-the-art performance on retrieval, both single and multi vector, across length
 - **Class of its own** for code
- **Lightspeed processing**
 - Very large gap for **long / variable lengths**

	Model	DPR		CoBERT		NLU	Code	
		BEIR	MLDR	BEIR	MLDR	GLUE	CSN	SQA
Base	RoBERTa	37.7	32.8	48.7	28.2	86.4	44.3	59.6
	DeBERTaV3	20.2	13.4	47.1	21.9	88.1	17.5	18.6
	GTE-en-MLM	41.4	44.4	48.2	69.3	85.6	44.9	71.4
	ModernBERT	41.6	44.0	51.3	80.2	88.4	56.4	73.6
Large	RoBERTa	41.4	36.1	49.8	28.8	88.9	47.3	68.1
	DeBERTaV3	25.6	19.2	46.7	23.0	91.4	21.2	19.7
	GTE-en-MLM	42.5	48.9	50.7	71.3	87.6	40.5	66.9
	ModernBERT	44.0	48.6	52.4	80.4	90.4	59.5	83.9



Ettin: encoder vs decoder

- Open data replication of ModernBERT

Category	Dataset	Pre-training		Mid-training		Decay Phase	
		Tokens (B)	%	Tokens (B)	%	Tokens (B)	%
News	CC News	7.3	0.4	—	—	—	—
Code	Starcoder	263.9	15.5	38.4	15.4	—	—
Code	Code_Repos	—	—	—	—	20.2	26.5
Crawl	CC Head	356.6	20.9	—	—	—	—
Crawl	DCLM	837.2	49.1	—	—	—	—
Crawl	DCLM (Dolmino)	—	—	175.5	70.4	26.0	34.1
Math	Open-Web-Math	12.7	0.7	—	—	—	—
Math	Algebraic StackExchange	12.6	0.7	—	—	—	—
Math	Math (Dolmino)	—	—	10.4	4.2	5.0	6.6
Scientific	PeS2o	57.3	3.4	8.3	3.3	—	—
Scientific	Arxiv	28.0	1.6	4.1	1.6	3.0	3.9
Social	Reddit	80.3	4.7	6.2	2.5	—	—
Social	StackExchange	19.6	1.1	—	—	—	—
Social	StackExchange (Dolmino)	—	—	2.7	1.1	4.0	5.2
Reference	Textbooks	—	—	—	—	0.5	0.7
Reference	Dolma Books	5.3	0.3	0.8	0.3	10.5	13.8
Reference	Wikipedia	7.3	0.4	0.5	0.2	3.0	3.9
Instruction	Tulu Flan	16.6	1.0	2.4	1.0	4.1	5.4
Total		1,704.7	100.0	249.3	100.0	76.3	100.0

Table 2: Training data mixture across the various training stages (pre-training, mid-training, decay). Later stages use higher quality data, from the recently released Dolmino dataset (OLMo et al., 2025). Dashes indicate that no data from that source was used. We trained for 1.7T tokens for pre-training, 250B for mid-training, and 50B for the decay phase. We sample from the dataset and repeat (or under-sample) as needed to hit the token counts used for training.

Ettin: encoder vs decoder

- Open data replication of ModernBERT
- Training of 6 sizes (from 17M to 1B) of encoder/decoder
 - Only difference is **masking and training objective**

Parameter	17M (XXS)	32M (XS)	68M (Small)	150M (Base)	400M (Large)	1B (XL)
Layers	7	10	19	22	28	28
Hidden Size	256	384	512	768	1024	1792
Intermediate Size	384	576	768	1152	2624	3840
Attention Heads	4	6	8	12	16	28
Learning Rate	3e-3	3e-3	3e-3	8e-4	5e-4	5e-4
Weight Decay	3e-4	3e-4	3e-4	1e-5	1e-5	5e-5
Warmup Tokens (B)	4	4	3	3	2	2
BS Warmup (B)	125	100	75	50	10	3

Table 1: Configuration for each Ettin model size. Both encoders and decoders use the exact same configuration, differing only in attention (bidirectional vs causal) and objective (MLM vs CLM).

Ettin: encoder vs decoder

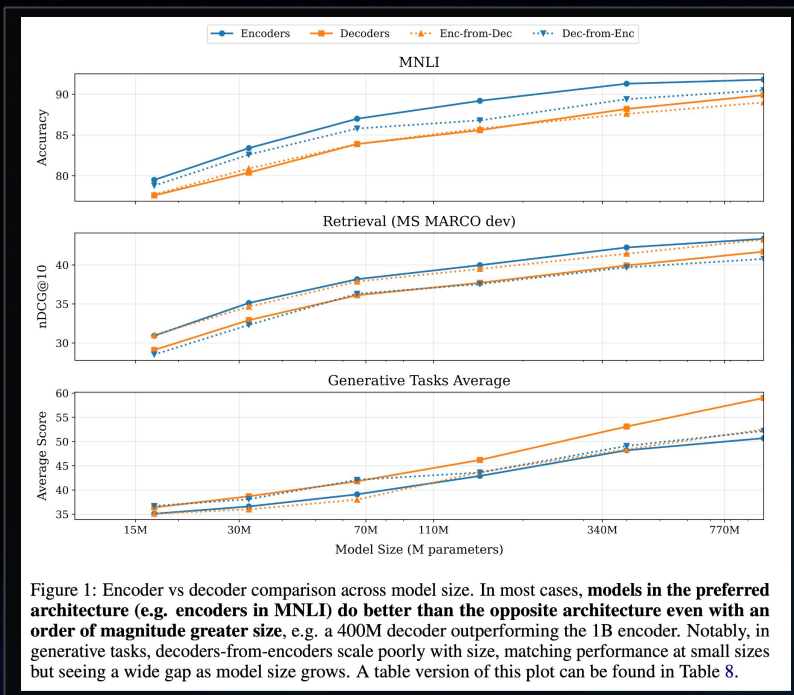
- Open data replication of ModernBERT
- Training of 6 sizes (from 17M to 1B) of encoder/decoder
 - Only difference is [masking](#) and [training objective](#)
- SOTA across sizes for both architecture*

Model Name	Embedding Tasks					GLUE Tasks		
	CSN	MLDR	Clustering	Retrieval	MTEB v2	SST-2	MNLI	GLUE Avg
XXS Models (7-17M parameters)								
BERT-mini	41.3	16.8	39.0	34.7	49.2	88.3	77.2	76.4
TinyBERT	39.8	14.2	37.4	33.3	49.7	91.2	80.9	77.0
Ettin-Enc-17m	59.1	24.4	39.1	35.6	48.9	91.2	79.5	79.2
XS Models (28-33M parameters)								
BERT-small	46.0	19.9	39.6	38.1	51.1	90.1	79.2	79.0
MiniLM L12	48.3	19.6	37.8	38.4	51.3	93.3	85.6	84.6
Ettin-Enc-32m	69.2	28.4	39.6	39.7	50.9	92.0	83.4	83.5
Small Models (68-82M parameters)								
DistilBERT	47.9	23.7	39.8	40.8	52.7	92.2	82.7	81.5
DistilRoBERTa	60.3	19.7	39.3	40.0	51.8	93.1	84.7	83.8
Ettin-Enc-68m	75.1	30.1	40.1	43.1	52.6	94.4	87.0	87.2
Base Models (123-150M parameters)								
BERT base	51.0	24.8	40.4	41.2	52.9	93.1	85.4	84.7
ModernBERT base	75.9	30.4	41.3	43.9	54.0	96.0	89.1	88.4
Ettin-Enc-150m	76.3	31.8	41.5	45.7	54.0	95.8	89.2	88.9
Large Models (353-395M parameters)								
BERT large	54.4	25.3	41.5	42.9	53.8	93.3	86.3	85.2
ModernBERT large	78.3	34.9	41.5	47.0	55.0	97.1	90.8	90.4
Ettin-Enc-400m	80.7	36.2	41.8	48.4	55.5	96.7	91.3	90.8
XL Models (884M-1.2B parameters)								
DeBERTa-v1-xl	75.6	28.1	42.5	47.2	56.4	97.1	91.7	90.7
Ettin-Enc-1B	82.3	40.2	41.9	50.1	56.0	97.1	91.8	91.6

Model Name	ARC	HS	LMB	OBQA	PIQA	SciQ	SIQA	TQA	WG	WSC	Avg
XXS Models (14-17M parameters)											
Pythia-14m	21.2	26.0	7.1	26.2	55.2	43.8	33.4	0.0	50.3	51.6	31.5
Ettin-Dec-17m	21.3	27.1	23.0	27.2	57.7	71.1	35.4	2.6	50.9	48.0	36.4
XS Models (32M parameters)											
Ettin-Dec-32m	23.5	28.5	28.5	28.2	57.7	77.5	36.4	3.8	53.1	50.2	38.7
Small Models (68-82M parameters)											
DistilGPT	23.0	27.5	25.0	26.8	59.8	62.6	36.1	0.3	50.4	53.8	36.5
Ettin-Dec-68m	25.3	33.4	35.2	29.4	61.8	83.2	38.8	5.6	50.1	55.3	41.8
Base Models (135-160M parameters)											
SmolLM2-135m	29.1	43.1	42.9	32.4	68.4	78.5	39.4	5.0	53.7	59.7	45.2
Pythia-160m	24.0	30.2	32.9	26.4	62.0	67.2	36.9	0.4	52.4	58.2	39.1
Ettin-Dec-150m	28.6	40.3	43.2	29.2	66.6	89.6	40.1	11.2	53.7	59.0	46.2
Large Models (360-410M parameters)											
Pythia-410m	24.7	40.6	51.5	29.4	67.0	72.3	39.0	1.8	53.6	65.2	44.5
SmolLM2-360m	37.6	56.3	53.5	37.6	71.8	86.6	40.7	18.4	58.6	70.3	53.1
Ettin-Dec-400m	33.6	54.3	52.3	34.4	71.0	91.8	45.5	18.3	57.6	71.8	53.1
XL Models (908M-1.2B parameters)											
OLMo-1B-0724	32.3	66.1	61.0	35.6	75.1	91.8	49.2	1.2	61.6	76.9	55.1
Llama-3.2-1B	36.2	63.7	62.1	37.2	75.0	88.4	43.2	24.9	60.6	74.7	56.6
Ettin-Dec-1B	39.7	62.9	58.4	41.6	74.4	93.8	48.2	29.3	62.7	79.1	59.0

Ettin: encoder vs decoder

- Comparison of encoder, decoder, encoder from decoder and decoder from encoder on generation, retrieval and classification
- Encoder >> decoder for discriminative tasks
 - Encoder from decoder **still lacks on classification, but close the gap for retrieval!**
- Decoder >> encoder for generative tasks



Multimodal encoders

- Same observation for multimodal retrieval
- Same backbone but much higher performance...?

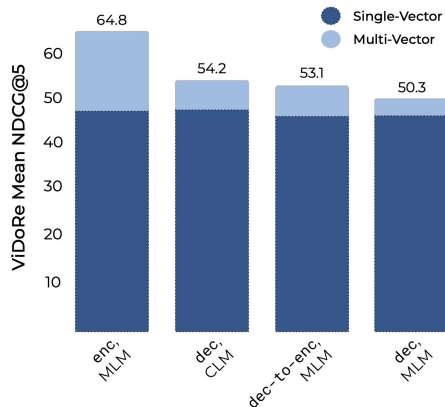
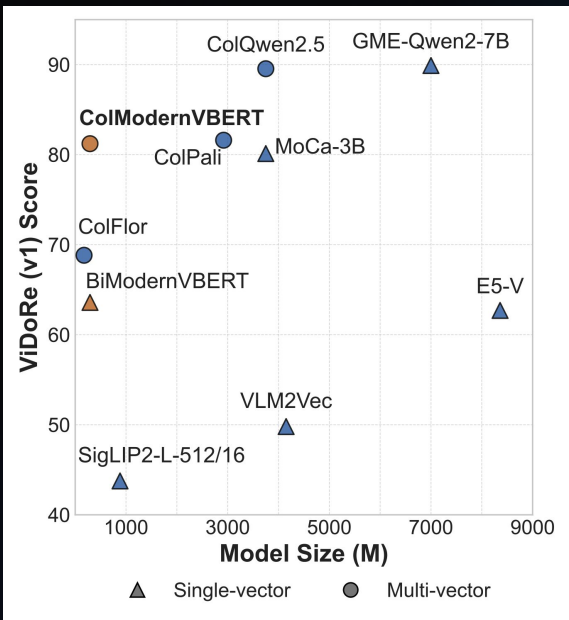


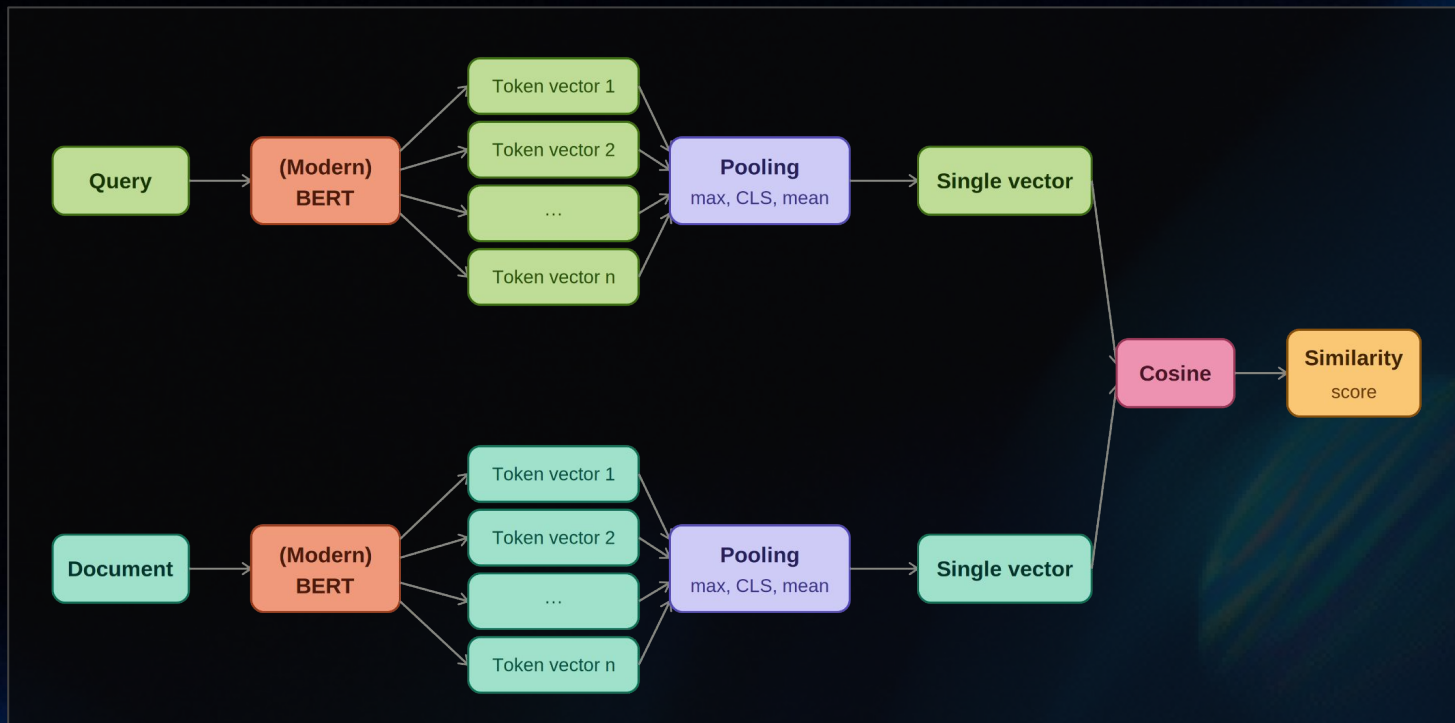
Figure 5: Impact of attention masks and training objectives on document retrieval performances. We report the average nDCG@5 on English splits of ViDoRe benchmarks for models post-trained on ColPali.



Multi-vector

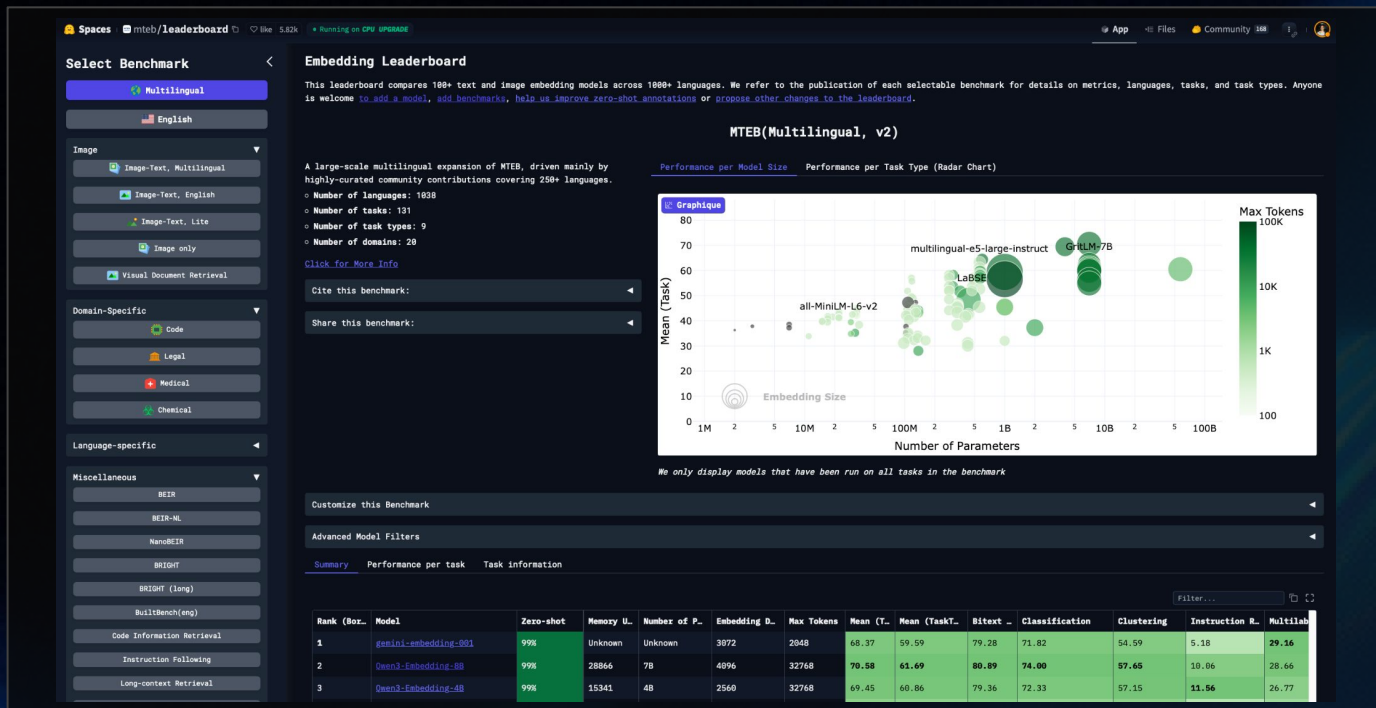
Dense (single) vector search

- Create contextualized representations of the tokens
- Pool them
- Compute cosine similarity



Performance evaluation

- Need evaluations to assess performance
- MTEB leaderboard centralizes various benchmarks and associated models performances
- Choose the model that **fits your budget and performs well on your domain**



BEIR: a school case of Goodhart's law

- Big challenge of ML is generalization: models can perform well in-domain but totally fail on data they have not been trained on
- BEIR introduced various benchmarks to evaluate out-of-domain performance
- Few years and thousands of models later: the hill has been climbed
- **Run your own evaluations!**

BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models

Nandan Thakur, Nils Reimers, Andreas Rücklé*, Abhishek Srivastava, Iryna Gurevych
Ubiquitous Knowledge Processing Lab (UKP-TUDA)
Department of Computer Science, Technische Universität Darmstadt
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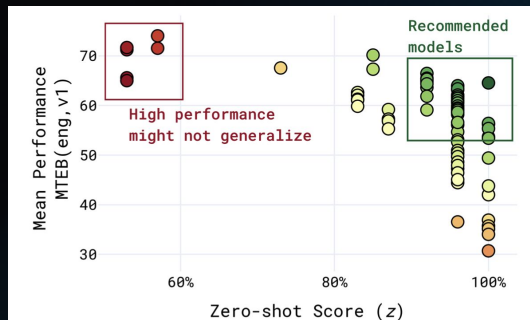
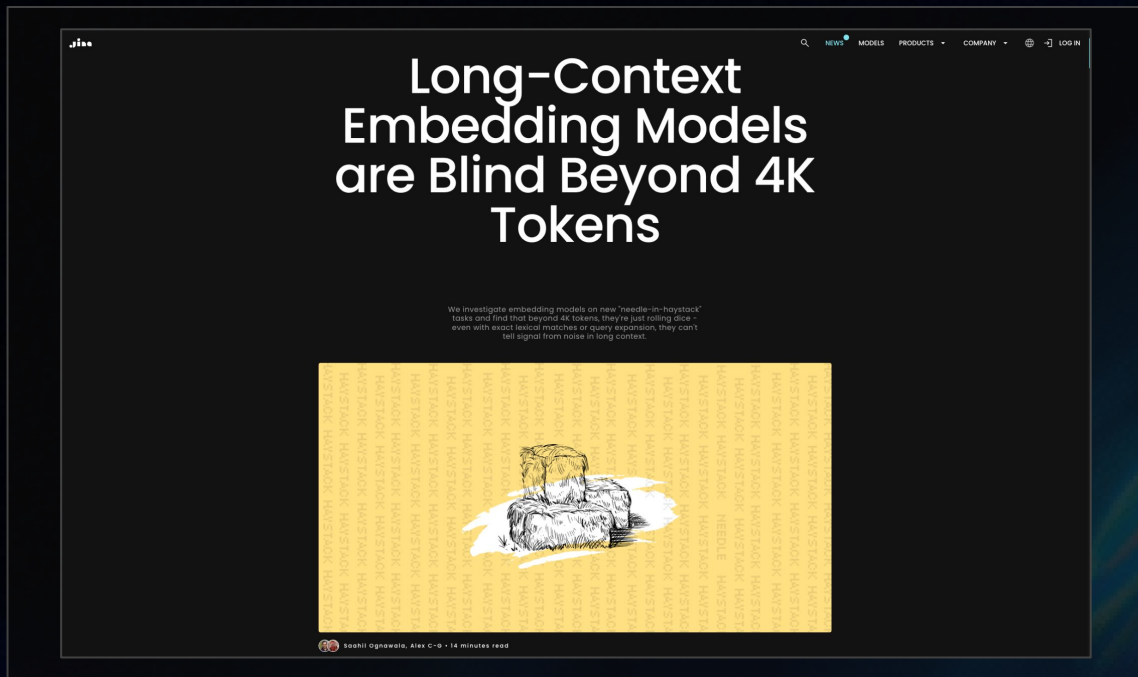


Figure 2. Models' mean performance against their zero-shot score on the legacy English MTEB. The highest ranking models achieve their scores by training on benchmark tasks, even though models with lower scores might generalize better to out-of-distribution environments.

If you cannot measure it, you cannot improve it

- Besides out-of-domain, different capabilities are not captured by existing benchmarks
- Long context: old models had context window of 512, but 8k models actually do not properly support 8k



If you cannot measure it, you cannot improve it

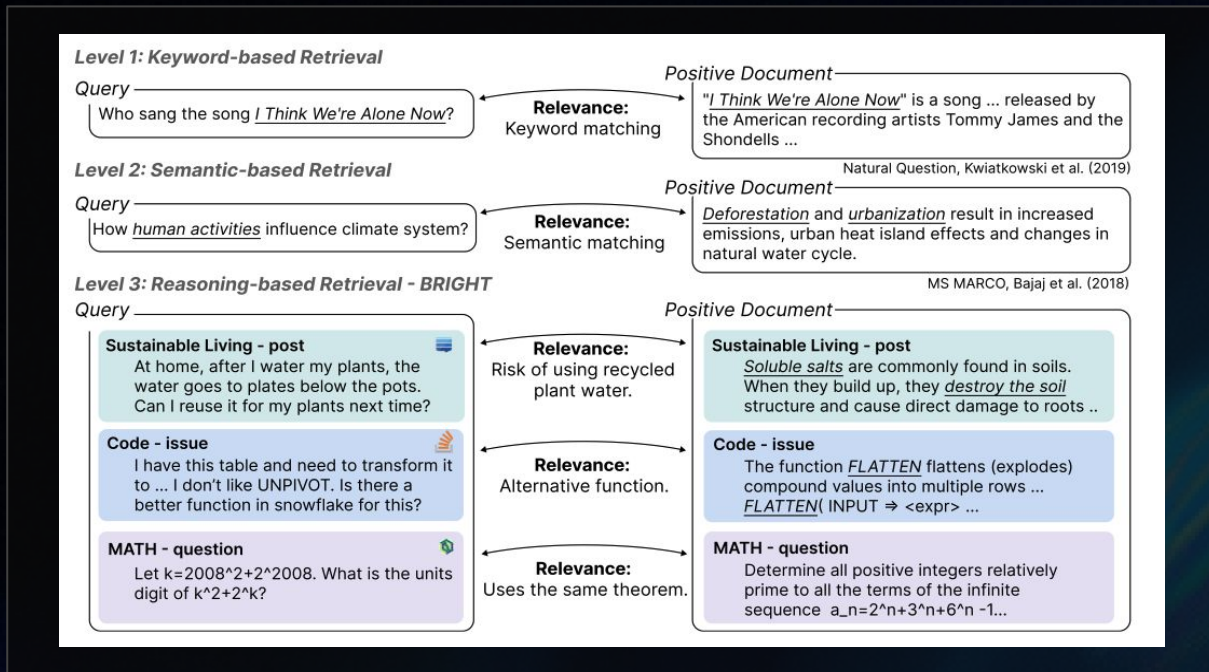
- Besides out-of-domain, different capabilities are not captured by existing benchmarks
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Model	Param.	CTX Len.	Synthetic (Acc@1)		Real (nDCG@10)				Avg.
			Passkey	Needle	NQA	QMS	SFD	WQA	
512 Context Models									
E5 _{Base} (Wang et al., 2022)	110M	512	38.0	28.5	25.3	23.8	74.7	55.8	41.0
E5-RoPE _{Base}	110M	512	38.5	31.5	24.6	23.2	66.6	58.8	40.5
GTE _{Base} (Li et al., 2023)	110M	512	31.0	24.5	28.6	21.8	55.8	47.3	34.8
BGE _{Base} (Xiao et al., 2023)	110M	512	18.0	25.3	25.6	22.4	60.3	51.7	33.9
Contriever (Izacard et al., 2021)	110M	512	38.5	29.0	26.7	25.5	73.5	47.3	40.1
GTR _{Base} (Ni et al., 2022)	110M	512	38.5	26.3	26.5	18.3	63.7	52.2	36.5
≥ 4k Context Models									
E5-Mistral (Wang et al., 2023b)	7B	4,096	71.0	48.3	44.6	43.6	96.8	82.0	64.4
Jina-V2 (Günther et al., 2023)	137M	8,192	50.3	54.5	37.9	38.9	93.5	74.0	58.2
Nomic-V1(Nussbaum et al., 2024)	137M	8,192	60.7	39.5	41.2	36.7	93.0	73.8	57.5
BGE-M3 (Chen et al., 2024)	568M	8,192	59.3	40.5	45.8	35.5	94.0	78.0	58.9
OpenAI-Ada-002	-	-	50.8	36.8	41.1	40.0	91.8	80.1	56.8
Our Extended Models									
E5 _{Base} + Tuning (4k)	110M	4,096	67.3	41.5	30.4	35.7	95.2	69.2	56.6
E5-RoPE _{Base} + SelfExtend (4k)	110M	4,096	73.5	53.5	32.3	39.1	91.9	74.6	60.8
E5-Mistral + NTK (32k)	7B	32,768	93.8	66.8	49.8	49.2	97.1	95.2	75.3

Table 2: Results (%) of existing and extended embedding models on LONGEMBED. *NQA*, *QMS*, *SFD*, *WQA* is short for *NarrativeQA*, *QMSum*, *SummScreenFD*, *2WikiMultihopQA*, respectively. We show that context window extension can effectively improve existing embedding models in processing long context.

If you cannot **measure it**, you cannot **improve it**

- Besides out-of-domain, different capabilities are not captured by existing benchmarks
- Long context: old models had context window of 512, but **8k models actually do not properly support 8k**
- Retrieval beyond keyword/semantic matching is challenging



If you cannot measure it, you cannot improve it

- Besides out-of-domain, different capabilities are not captured by existing benchmarks
- Long context: old models had context window of 512, but **8k models actually do not properly support 8k**
- Retrieval beyond keyword/semantic matching is challenging

Table 2: **The performance of retrieval models on BRIGHT.** We report nDCG@10 for all datasets: Biology (Bio.), Earth Science (Earth.), Economics (Econ.), Psychology (Psy.), Robotics (Rob.), Stack Overflow (Stack.), Sustainable Living (Sus.), LeetCode (Leet.), Pony, AoPS, TheoremQA with question retrieval (TheoQ.) and with theorem retrieval (TheoT.). Avg. denotes the average score across 12 datasets. The best score on each dataset is shown in bold and the second best is underlined. We show that reasoning-intensive retrieval is challenging for current retrievers, where the best model only achieves an nDCG@10 score of 24.3 on average. Model details are in Appendix A.1

	StackExchange							Coding		Theorem-based			Avg.
	Bio.	Earth.	Econ.	Psy.	Rob.	Stack.	Sus.	Leet.	Pony	AoPS	TheoQ.	TheoT.	
Sparse model													
BM25	18.9	27.2	14.9	12.5	13.6	18.4	15.0	24.4	7.9	6.2	10.4	4.9	14.5
Open-sourced models (<1B)													
BGE	11.7	24.6	16.6	17.5	11.7	10.8	13.3	26.7	5.7	6.0	13.0	6.9	13.7
Inst-L	15.2	21.2	14.7	22.3	11.4	13.3	13.5	19.5	1.3	8.1	20.9	9.1	14.2
SBERT	15.1	20.4	16.6	22.7	8.2	11.0	15.3	26.4	7.0	5.3	20.0	10.8	14.9
Open-sourced models (>1B)													
E5	18.6	26.0	15.5	15.8	16.3	11.2	18.1	28.7	4.9	7.1	26.1	26.8	17.9
SFR	19.1	26.7	17.8	19.0	16.3	14.4	19.2	27.4	2.0	7.4	24.3	26.0	18.3
Inst-XL	21.6	34.3	22.4	27.4	18.2	21.2	19.1	27.5	5.0	8.5	15.6	5.9	18.9
GritLM	24.8	32.3	18.9	19.8	17.1	13.6	17.8	29.9	22.0	8.8	25.2	21.2	21.0
Qwen	30.6	36.4	17.8	24.6	13.2	22.2	14.8	25.5	9.9	14.4	27.8	32.9	22.5
Proprietary models													
Cohere	18.7	28.4	20.4	21.6	16.3	18.3	17.6	26.8	1.9	6.3	15.7	7.2	16.6
OpenAI	23.3	26.7	19.5	27.6	12.8	14.3	20.5	23.6	2.4	8.5	23.5	11.7	17.9
Voyage	23.1	25.4	19.9	24.9	10.8	16.8	15.4	30.6	1.5	7.5	27.4	11.6	17.9
Google	22.7	34.8	19.6	27.8	15.7	20.1	17.1	29.6	3.6	9.3	23.8	15.9	20.0

If you cannot measure it, you cannot improve it

- Besides out-of-domain, different capabilities are not captured by existing benchmarks
- Long context: old models had context window of 512, but 8k models actually do not properly support 8k
- Retrieval beyond keyword/semantic matching is challenging
- BM25 (lexical search) is surprisingly strong on those challenges

Method	Synthetic		Real				Avg.
	P	N	NQA	QMS	SFD	WQA	
BM25	100	95.3	71.5	81.3	97.6	96.5	90.4
E5-Mistral	71.0	48.3	44.6	43.6	96.8	82.0	64.4
+NTK (32k)	93.8	66.8	49.8	49.2	97.1	95.2	75.3

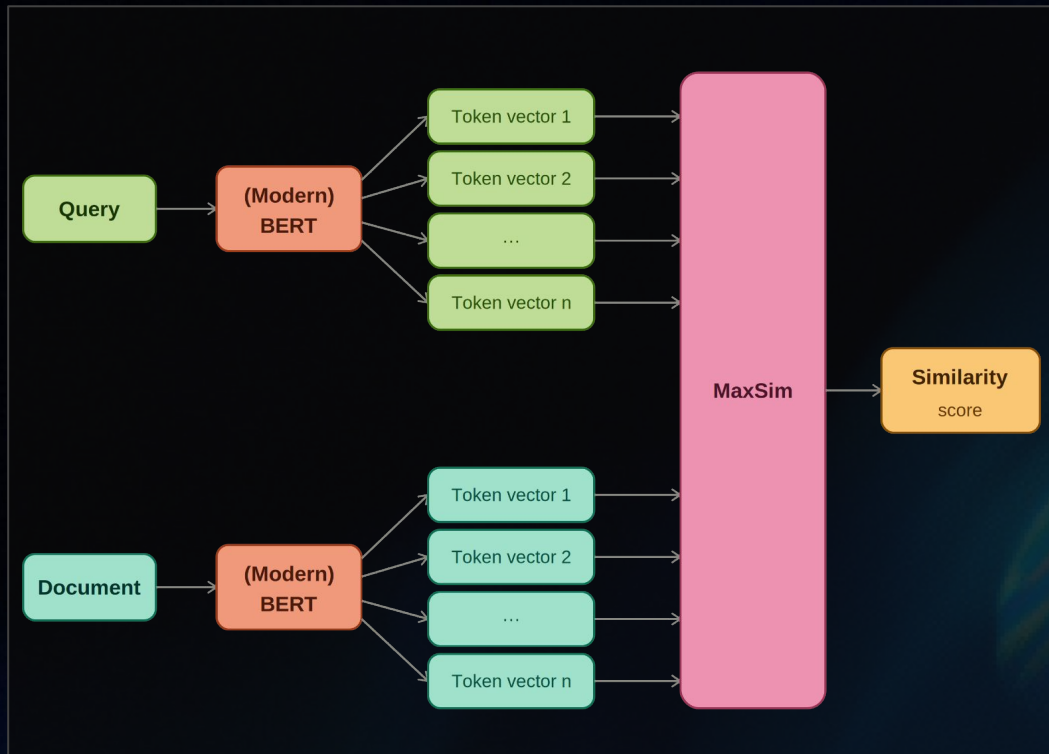
Performance of BM25 on the LongEmbed benchmark

	StackExchange							Coding		Theorem-based			Avg.
	Bio.	Earth.	Econ.	Psy.	Rob.	Stack.	Sus.	Leet.	Pony	AoPS	TheoQ.	TheoT.	
Sparse model													
BM25	18.9	27.2	14.9	12.5	13.6	18.4	15.0	24.4	7.9	6.2	10.4	4.9	14.5
Open-sourced models (<1B)													
BGE	11.7	24.6	16.6	17.5	11.7	10.8	13.3	26.7	5.7	6.0	13.0	6.9	13.7
Inst-L	15.2	21.2	14.7	22.3	11.4	13.3	13.5	19.5	1.3	8.1	20.9	9.1	14.2
SBERT	15.1	20.4	16.6	22.7	8.2	11.0	15.3	26.4	7.0	5.3	20.0	10.8	14.9
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E5	18.6	26.0	15.5	15.8	16.3	11.2	18.1	28.7	4.9	7.1	26.1	26.8	17.9
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Inst-XL	21.6	34.3	22.4	27.4	18.2	21.2	19.1	27.5	5.0	8.5	15.6	5.9	18.9
GritLM	24.8	32.3	18.9	19.8	17.1	13.6	17.8	29.9	22.0	8.8	25.2	21.2	21.0
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Proprietary models													
Cohere	18.7	28.4	20.4	21.6	16.3	18.3	17.6	26.8	1.9	6.3	15.7	7.2	16.6
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Voyage	23.1	25.4	19.9	24.9	10.8	16.8	15.4	30.6	1.5	7.5	27.4	11.6	17.9
Google	22.7	34.8	19.6	27.8	15.7	20.1	17.1	29.6	3.6	9.3	23.8	15.9	20.0

Performance of BM25 on the BRIGHT benchmark

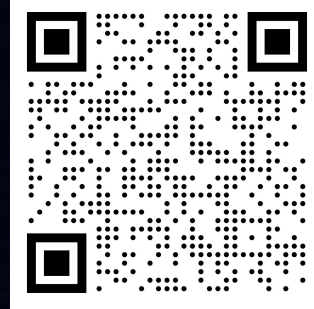
Replacing pooling with late interaction

- Do not pool the vectors; **keep all the information!**
- Use a similarity operator working at the token level



(Why it is very different)

- Not enough time to cover the *why* it is an important difference
- Focusing on the *practical* aspect



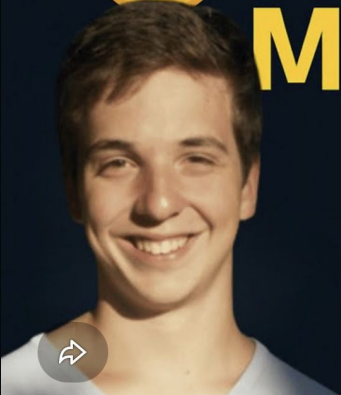
Late Interaction Beats Single Vectors For RAG, introduction to PyLate
Hamel Husain

Single Vector?

 **YOU'RE MISSING OUT**



RAG
Regarder sur YouTube



A simple, yet effective, difference

- Strong performance out-of-domain, outperforming in-domain dense models
- Out-of-domain is tough to define, feedback from practitioners
 - Test on [your own data!](#)

Our model prior to the post-training step, JaColBERTv2.4, also outperforms all other existing approaches, even while being fully out-of-domain on every evaluation dataset.

similarity tasks. On 22 of 28 out-of-domain tests, ColBERTv2 achieves the highest quality, outperforming the next best retriever by up to 8% relative gain, while using its compressed representations.

of our XMOD-based dense retrievers. We observe that ColBERT-XM surpasses DPR-XM in the training language (i.e., English) by 4.5% on MRR@10. Furthermore, it consistently outperforms DPR-XM across the other 13 languages not encountered during training by an average of 4.9%. Supported by findings from [Santhanam et al. \(2022\)](#), our results confirm that multi-vector models bypass the restrictive information bottleneck inherent in single-vector models, enabling a richer and more nuanced representation of queries and passages, thereby yielding higher retrieval performance.

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Table 4. MAP and P@10 on CLEF Title and Title+Description queries. Bold are best among a year; italics are best in a row (*i.e.*, with and without neural machine translation), † indicates significant difference from BM25+ITD by paired *t*-test with 16-test Bonferroni correction ($p < 0.05$).

Query Set	ITD	MAP						P@10					
		MULM	BM25	ColBERT-X		DPR-X		MULM	BM25	ColBERT-X		DPR-X	
				MTT-M	ET	MTT-M	ET			MTT-M	ET	MTT-M	ET
Title Queries													
2001	✓	–	<i>0.398</i>	0.377	0.391	0.338	0.344	–	<i>0.648</i>	0.612	0.596	0.548	0.584
	✗	0.349	–	<i>0.360</i>	0.322	0.327	0.298†	0.650	–	0.600	0.588	0.592	0.570
2002	✓	–	0.337	0.367	0.389	0.287	0.304	–	0.618	0.606	0.670	0.530	0.596
	✗	0.276	–	0.352	0.333	0.282	0.277	0.592	–	0.614	<i>0.622</i>	0.544	0.556
2003	✓	–	<i>0.349</i>	0.337	0.349	0.276†	0.266†	–	0.595	0.542	0.573	0.517	0.497†
	✗	0.305	–	0.332	0.290	0.273†	0.247†	0.497	–	<i>0.546</i>	0.541	0.527	0.492†
All	✓	–	0.361	0.359	0.375	0.299†	0.302†	–	0.619	0.583	0.611	0.531†	0.554†
	✗	0.310	–	<i>0.347</i>	0.314†	0.293†	0.273†	0.575	–	<i>0.584</i>	0.581	0.553†	0.536†
Title + Description Queries													
2001	✓	–	0.436	0.472	0.477	0.365	0.356	–	0.704	0.718	0.754	0.658	0.650
	✗	0.387	–	<i>0.462</i>	0.405	0.358	0.324†	0.700	–	0.704	<i>0.744</i>	0.658	0.644
2002	✓	–	0.398	0.470†	0.480 †	0.347	0.332	–	0.696	0.774	0.770	0.664	0.620
	✗	0.347	–	<i>0.462</i>	0.410	0.335	0.310	0.666	–	<i>0.752</i>	0.720	0.672	0.640
2003	✓	–	0.394	0.419	0.410	0.343	0.328†	–	0.615	0.646	0.661	0.620	0.600
	✗	0.376	–	<i>0.409</i>	0.358	0.338	0.302†	0.563	–	<i>0.653</i>	0.637	0.622	0.575
All	✓	–	0.408	0.451†	0.453 †	0.351†	0.338†	–	0.669	0.709	0.725 †	0.646	0.622
	✗	0.368	–	<i>0.442</i>	0.390	0.343†	0.312†	0.643	–	<i>0.700</i>	<i>0.697</i>	0.639	0.617

A simple, yet effective, difference

- GTE-ModernColBERT is SOTA on the Long Embed by a large margin while being trained on document length of 300
 - Performance improves even from 8k -> 16k while even the base model has been trained at 8k max

Model	Mean
GTE-ModernColBERT	88.39
voyage-multilingual-2	79.17
snowflake-arctic-embed-l-v2,0	63.73
gte-multilingual-base	62.12
bge-m3	58.73

Performance (NDCG@10) of different models on the Long Embed benchmark

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- Reason-ModernColBERT outperforms all 7B models (45× bigger) on BRIGTH, competitive with ReasonIR-8B trained on the same data

Model / Metric	Full mean
BM25	14.53
< 1B OS	
BGE	13.71
Inst-L	14.21
SBERT	14.9
> 1B OS	
E5	17.93
SFR	18.3
Inst-XL	18.89
GritLM	20.95
Qwen	22.51
Proprietary	
Cohere	16.6
OpenAI	17.87
Voyage	17.91
Google	20
ReasonIR data	
ReasonIR-8B	24.38
Reason-ModernColBERT (150M)	22.62

Performance (NDCG@10) of different models on the BRIGTH benchmark

A simple, yet effective, difference

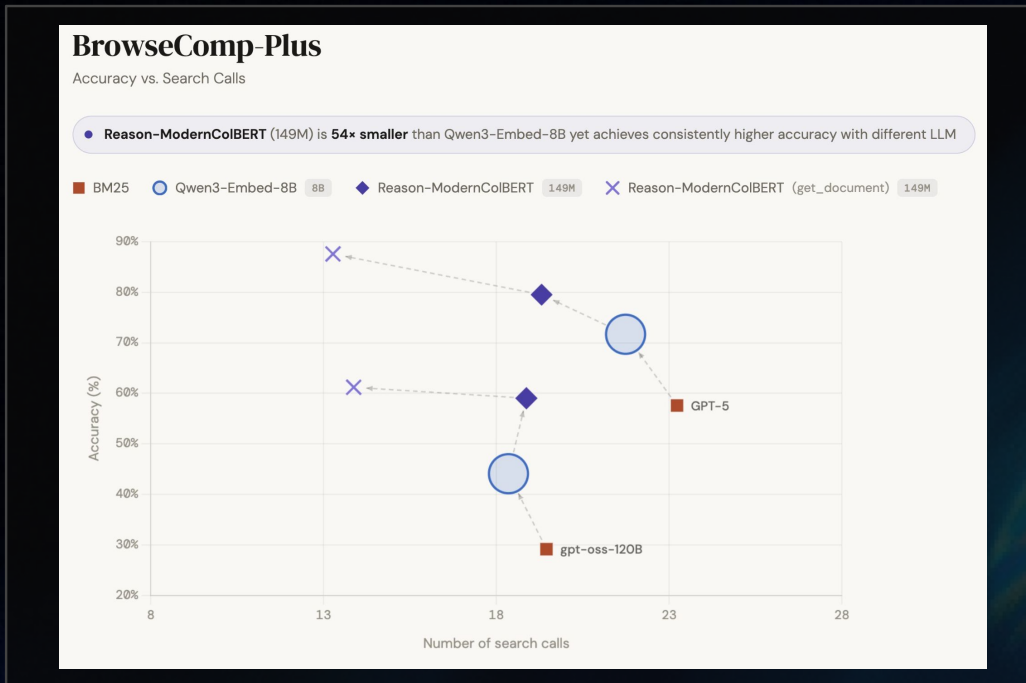
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- Reason-ModernColBERT outperforms all 7B models (45× bigger) on BRIGHT, competitive with ReasonIR-8B trained on the same data

Model	Mean
Dense (single vector) model	12.31
Late-interaction (multi vector model)	19.61

Comparison of single and multi vector models on the BRIGHT benchmark

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 - Outperforms Qwen-8B (52× bigger) on BrowseComp-Plus



Accuracy vs number of search calls for the BrowseComp-Plus benchmark

A simple, yet effective, difference

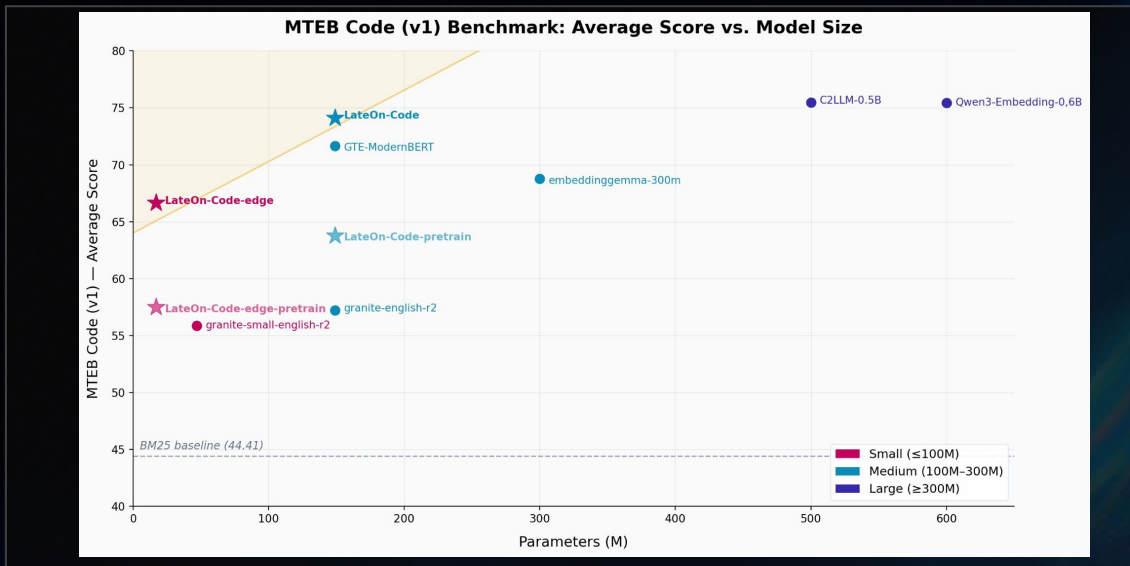
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#	LLM	RETRIEVER	SUBMITTED BY	SCAFFOLD	ACCURACY (%)	RECALL (%)	SEARCH CALLS	CALIBRATION ERROR (%)
1	GPT-5 PROPRIETARY Mar 16, 2026	Reason-ModernColBERT	LightOn	Custom	87.59%	81.55%	13.27	6.07%
2	GPT-5-mini PROPRIETARY Nov 26, 2025	openJiuwen-deepsearch	openJiuwen	Custom	80.00%	54.51%	6.05	18.82%
3	GPT-5 PROPRIETARY Mar 16, 2026	Reason-ModernColBERT	LightOn	Standard	79.52%	83.52%	19.31	7.46%
4	GPT-5 PROPRIETARY Dec 11, 2025	Mixedbread Search	Agentica	Custom	78.41%	48.85%	44.67	7.92%
5	GPT-5 PROPRIETARY Oct 22, 2025	Mixedbread Search	Mixedbread	Standard	74.10%	80.29%	21.22	11.00%
6	GPT-5 PROPRIETARY Aug 8, 2025	Qwen3-Embed-8B	BrowseComp-Plus	Standard	71.69%	78.98%	21.74	9.58%

BrowseComp-Plus leaderboard

A simple, yet effective, difference

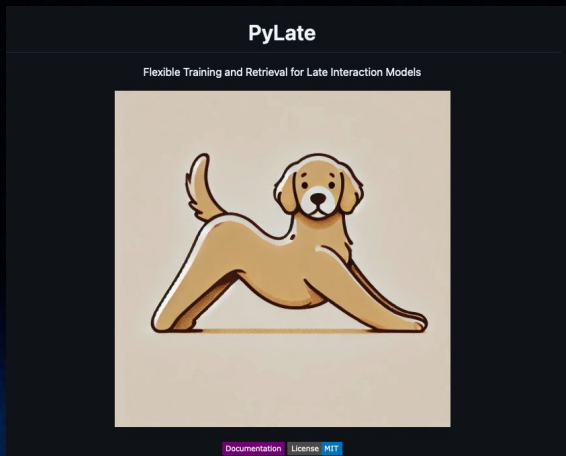
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- Reason-ModernColBERT outperforms all 7B models (45× bigger) on BRIGHT, competitive with ReasonIR-8B trained on the same data
 - Outperforms Qwen-8B (52× bigger) on BrowseComp-Plus. Actually sets a new SOTA on the bench
- LateOn-Code-edge (17M) outperforms all models < 150M, close to Gemma. LateOn-Code is very close to LLM-based models
 - Plugged into ColGrep, runs on your laptop at lightning speed, enhance Claude Code and save a lot of \$



MTEB Code (v1) performance against size

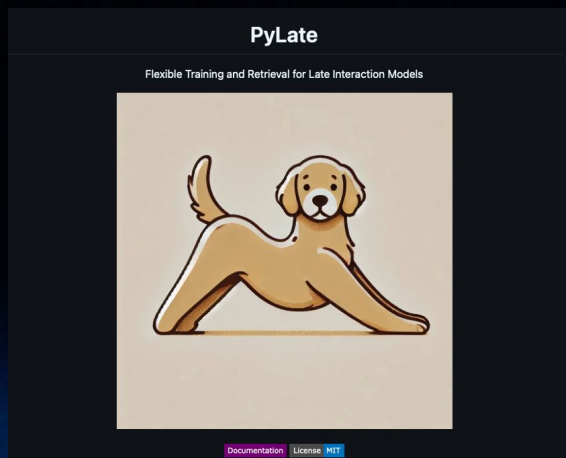
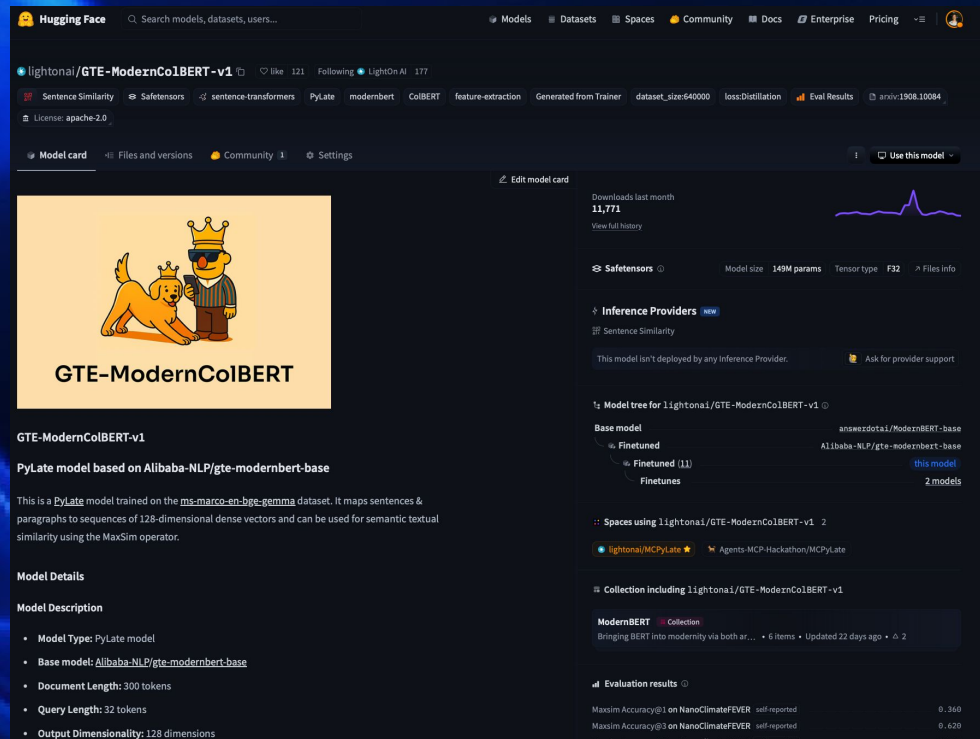
PyLate: extending Sentence Transformers for multi-vector models

- Late interaction = dense - pooling + MaxSim
- Building upon ST allows:
 - Support of **all base models**
 - **Efficient** (multi-GPU, FP/BF16, gradient checkpointing, ...) and **monitorable** (W&B) training



PyLate: extending Sentence Transformers for multi-vector models

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 - Integration in the **HF ecosystem** (automatic model card creation, push to hub, ...)

Hugging Face Search models, datasets, users...

Models Datasets Spaces Community Docs Enterprise Pricing

lightonai/GTE-ModernColBERT-v1 Like 121 Following LightOn AI 177

Sentence Similarity Safetensors sentence-transformers PyLate modernbert ColBERT feature-extraction Generated from Trainer dataset_size:640000 loss:Distillation Eval Results arxiv:1906.10094

License: apache-2.0

Model card Files and versions Community Settings Use this model

GTE-ModernColBERT

GTE-ModernColBERT-v1

PyLate model based on Alibaba-NLP/gte-modernbert-base

This is a **PyLate** model trained on the [ms-marco-en-bpe-gemma](#) dataset. It maps sentences & paragraphs to sequences of 128-dimensional dense vectors and can be used for semantic textual similarity using the MaxSim operator.

Model Details

Model Description

- **Model Type:** PyLate model
- **Base model:** [Alibaba-NLP/gte-modernbert-base](#)
- **Document Length:** 300 tokens
- **Query Length:** 32 tokens
- **Output Dimensionality:** 128 dimensions

Downloads last month
11,771
[View full history](#)

Safetensors Model size: 149M params Tensor type: F32 Files info

Inference Providers NEW

Sentence Similarity

This model isn't deployed by any Inference Provider. [Ask for provider support](#)

Model tree for lightonai/GTE-ModernColBERT-v1

Base model [answercdotai/ModernBERT-base](#)

Finetuned [Alibaba-NLP/gte-modernbert-base](#) [this model](#)

Finetunes 2 models

Spaces using lightonai/GTE-ModernColBERT-v1

[lightonai/MCPyLate](#) [Agents-MCP-Hackathon/MCPyLate](#)

Collection including lightonai/GTE-ModernColBERT-v1

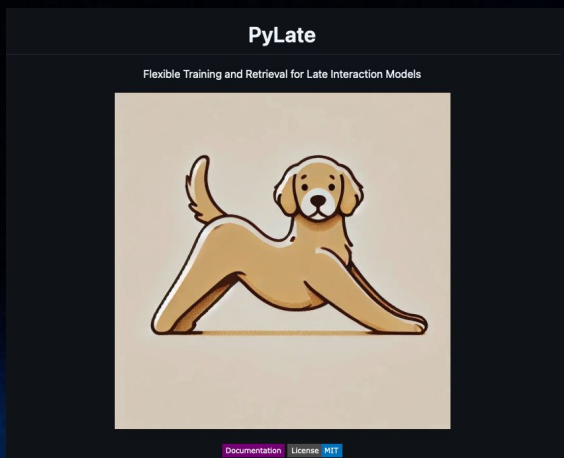
ModernBERT [Collection](#)
Bringing BERT into modernity via both ar... • 6 items • Updated 22 days ago • 2

Evaluation results

Model	Task	Score
MaxSim Accuracy@1	on NanoClimateFEVER	self-reported 0.360
MaxSim Accuracy@3	on NanoClimateFEVER	self-reported 0.620

PyLate: extending Sentence Transformers for multi-vector models

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- Building upon ST allows:
 - Support of **all base models**
 - **Efficient** (multi-GPU, FP/BF16, gradient checkpointing, ...) and **monitorable** (W&B) training
 - Integration in the **HF ecosystem** (automatic model card creation, push to hub, ...)
 - Very **familiar syntax**!



```

1 from datasets import load_dataset
2 from sentence_transformers import (
3     SentenceTransformerTrainer,
4     SentenceTransformerTrainingArguments,
5 )
6
7 from pylate import evaluation, losses, models, utils
8
9 # Define model parameters for contrastive training
10 model_name = "answerdotai/ModernBERT-base" # Choose the pre-trained model you want to use as base
11 batch_size = 32 # Larger batch size often improves results, but requires more memory
12
13 num_train_epochs = 1 # Adjust based on your requirements
14 # Set the run name for logging and output directory
15 run_name = "contrastive-bert-base-uncased"
16 output_dir = f"output/{run_name}"
17
18 # 1. Here we define our ColBERT model. If not a ColBERT model, will add a linear layer to the base encoder.
19 model = models.ColBERT(model_name_or_path=model_name)
20
21 # Load dataset
22 dataset = load_dataset("sentence-transformers/msmarco-bm25", "triplet", split="train")
23 # Split the dataset (this dataset does not have a validation set, so we split the training set)
24 splits = dataset.train_test_split(test_size=0.01)
25 train_dataset = splits["train"]
26 eval_dataset = splits["test"]
27
28 # Define the loss function
29 train_loss = losses.Contrastive(model=model)
30
31 # Initialize the evaluator
32 dev_evaluator = evaluation.NanoBEIREvaluator()
33
34 # Configure the training arguments (e.g., batch size, evaluation strategy, logging steps)
35 args = SentenceTransformerTrainingArguments(
36     output_dir=output_dir,
37     num_train_epochs=num_train_epochs,
38     per_device_train_batch_size=batch_size,
39     per_device_eval_batch_size=batch_size,
40     fp16=True, # Set to False if you get an error that your GPU can't run on FP16
41     bf16=False, # Set to True if you have a GPU that supports BF16
42     run_name=run_name, # Will be used in W&B if 'wandb' is installed
43     learning_rate=3e-6,
44 )
45
46 # Initialize the trainer for the contrastive training
47 trainer = SentenceTransformerTrainer(
48     model=model,
49     args=args,
50     train_dataset=train_dataset,
51     eval_dataset=eval_dataset,
52     loss=train_loss,
53     evaluator=dev_evaluator,
54     data_collator=utils.ColBERTCollator(model.tokenize),
55 )
56 # Start the training process
57 trainer.train()
58

```

Training is cool, show me the **evals**

- Built-in efficient PLAID indexes to easily and efficiently perform retrieval
- Helper functions to evaluate the output using **ranx**
- Standard **queries/documents/qrels** format
 - MTEB
 - ir-datasets
 - BEIR
- In-training evaluator
- Integrated in MTEB & FreshStack!

```

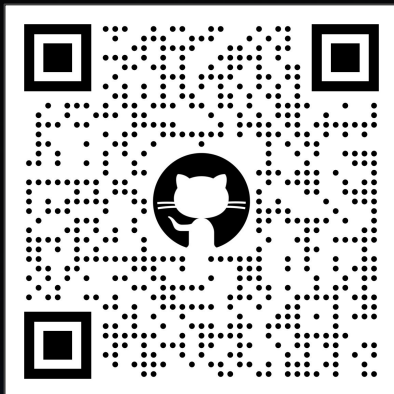
1 """Evaluation script for BEIR datasets with a PLAID index."""
2
3 import argparse
4
5 from pylatte import evaluation, indexes, models, retrieve
6
7 if __name__ == "__main__":
8     # Parse dataset_name from command line arguments
9     parser = argparse.ArgumentParser(description="Dataset name")
10    parser.add_argument(
11        "--dataset_name",
12        type=str,
13        default="nfcrcpus",
14        help="Name of the dataset to evaluate on (default: 'nfcrcpus')",
15    )
16    args = parser.parse_args()
17    dataset_name = args.dataset_name
18    model_name = "lightonai/GTE-ModernColBERT-v1"
19    model = models.ColBERT(
20        model_name_or_path=model_name,
21    )
22
23    documents, queries, qrels = evaluation.load_beir(
24        dataset_name=dataset_name, split="test"
25    )
26
27    index = indexes.PLAID(
28        index_name=f"{dataset_name}_{model_name.split('/')[0]}",
29    )
30
31    retriever = retrieve.ColBERT(index=index)
32
33    documents_embeddings = model.encode(
34        sentences=[document["text"] for document in documents],
35        batch_size=2000,
36        is_query=False,
37        show_progress_bar=True,
38    )
39
40    index.add_documents(
41        documents_ids=[document["id"] for document in documents],
42        documents_embeddings=documents_embeddings,
43    )
44    queries_embeddings = model.encode(
45        sentences=list(queries.values()),
46        is_query=True,
47        show_progress_bar=True,
48        batch_size=32,
49    )
50
51    scores = retriever.retrieve(queries_embeddings=queries_embeddings, k=100)
52
53    evaluation_scores = evaluation.evaluate(
54        scores=scores,
55        qrels=qrels,
56        queries=list(queries.keys()),
57        metrics=["map", "ndcg@10", "ndcg@100", "recall@10", "recall@100"],
58    )
59
60    print(evaluation_scores)
61

```

Conclusion

- Encoders are very strong for **discriminative tasks**
 - ModernBERT became a new standard for the community
 - Etnin (small) models are widely used and gave proper comparison ground
- Late interaction models exhibit very strong results
- They are well-suited to your **modern real-world use cases** (out-of-domain generalization, long context, reasoning-intensive, deep research, code retrieval ...)
- PyLate allows to easily train, experiment and evaluate those models with an experience very similar to sentence transformers
 - Go try existing models on your data and train your own specialized models!

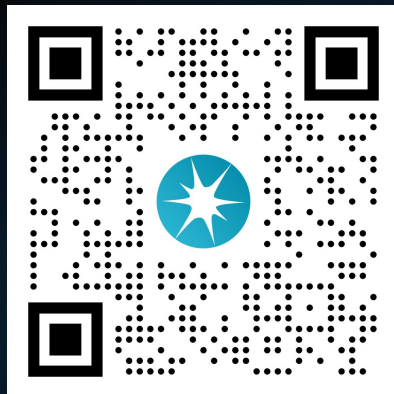
Get started with **PyLate**



Explore all our **models**



Contact us



@antoine_chaffin

Director's cut



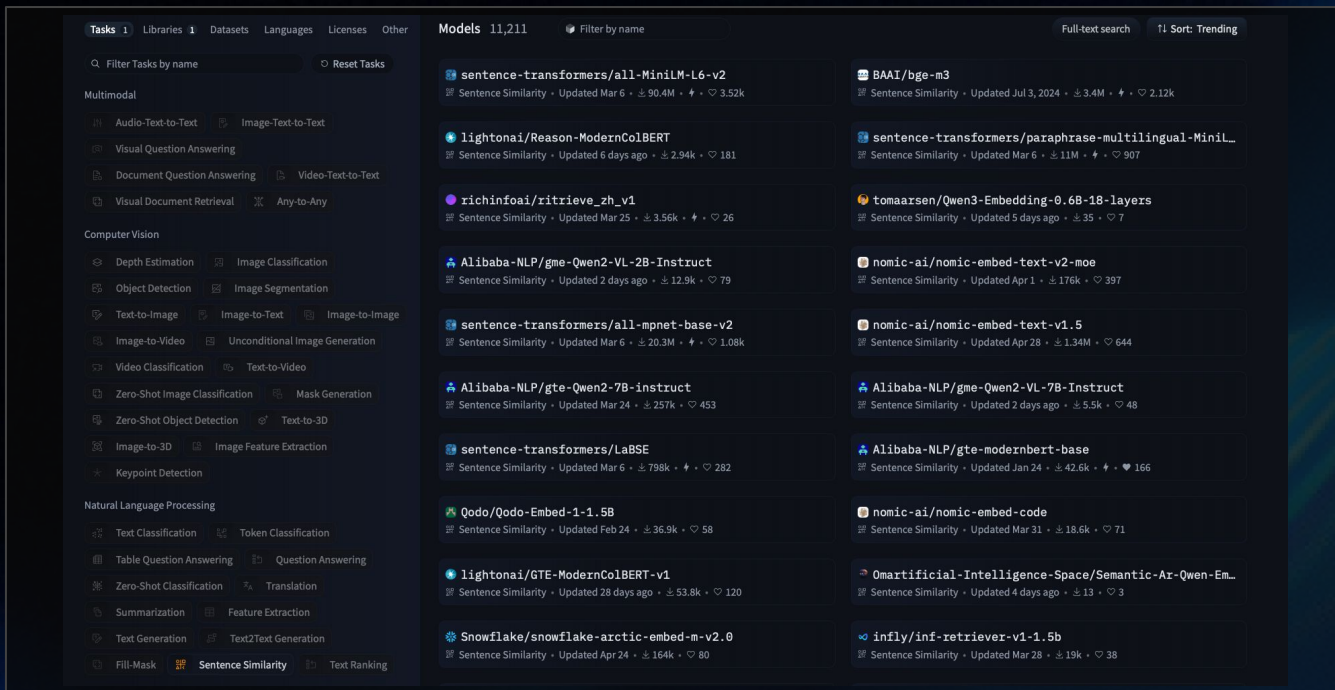
About Me

- R&D engineer at LightOn
- PhD on [multimodal misinformation detection](#)
 - Multimodal information retrieval
 - Guiding generative models for synthetic data generation
- Working (multimodal) information retrieval, encoders and late interaction
- Co-creator of [ModernBERT](#) & [PyLate](#)
- Designing [OCR-free RAG pipelines](#) (including MonoQwen reranker)
- Spending quite some time doing propaganda (and shitposting) on these subjects on Twitter ([@antoine_chaffin](#))



The go-to for RAG pipelines

- Great performance out of the box
- Lots (and lots) of available models
 - Sizes, languages, domains, ...
- Easy to deploy
 - VectorDBs, serving APIs



Pooling: the **intrinsic flaw** of dense models

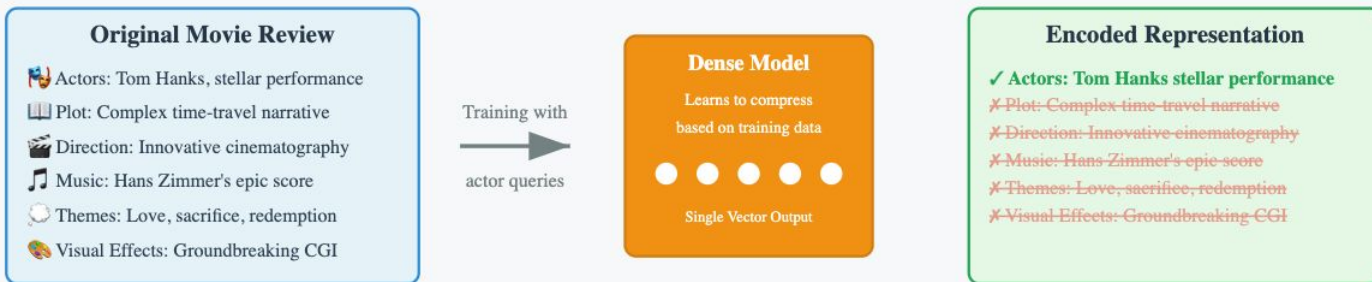
- Pooling operation **compresses** n tokens into a single one



Pooling: the intrinsic flaw of dense models

- Pooling operation compresses n tokens into a single one
- Selective behavior is learned during training (mainly through data)
- Gets more extreme with longer context
- Learn one notion of similarity

Dense Models Learn Selective Information Encoding



Problems in Production



Pooling: the **intrinsic flaw** of dense models

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BM25: Term Frequency + Inverse Document Frequency

Query: "Leonardo DiCaprio disaster"

"Titanic features **Leonardo** **DiCaprio** in a romantic **disaster** film..."

✓ **No pooling/compression**
✗ **Exact matching**

This is **not** just a big single vector

- If it is a compression issue, I can just use a big embedding vector right?



This is **not** just a big single vector

- If it is a compression issue, I can just use a big embedding vector right?
- Not quite, the different “views” of a document will conflict into this single representation
- MaxSim only select one document token per query token
 - Only modify the document tokens that *matched* the query
- Can fall back to “soft lexical matching” (learned BM25)

Dense (Single Vector)

"Titanic is a masterpiece featuring **Leonardo DiCaprio** and **Kate Winslet**. The **romantic storyline** between Jack and Rose unfolds as the **ship encounters disaster**. Their **forbidden love** drives the narrative."

Query: "Leonardo DiCaprio movies"

Query: "romantic disaster films"

↓ POOLING ↓

Single Compressed Vector

CONFLICTED REPRESENTATION

All information forced into one representation

⚠ **Actor and plot signals interfere with each other**

Late Interaction

"Titanic is a masterpiece featuring **Leonardo DiCaprio** and **Kate Winslet**. The **romantic storyline** between Jack and Rose unfolds as the **ship encounters disaster**. Their **forbidden love** drives the narrative."

Query: "Leonardo DiCaprio movies"

Query: "romantic disaster films"

↓ NO POOLING ↓

Token-Level Representations

Titanic	is	masterpiece	Leonardo	DiCaprio	Kate
Winslet	romantic	storyline	ship	disaster	love

MaxSim Operation:

- **Actor query** matches specific actor tokens
- **Plot query** matches specific plot tokens
- Each aspect gets clean, uninterrupted signal

A simple, yet effective, difference

- GTE-ModernColBERT is SOTA on the Long Embed by a large margin while being trained on document length of 300
 - Performance improves even from 8k -> 16k while even the base model has been trained at 8k max

Model	Mean	LEMBNarrativeQARetrieval	LEMBNeedleRetrieval	LEMBPaskeyRetrieval	LEMBQMSumRetrieval	LEMBSummScreenFDRetrieval	LEMBWikimQARetrieval
GTE-ModernColBERT	88.39	78.82	92.5	92	72.17	94.98	99.87
voyage-multilingual-2	79.17	64.694	75.25	97	51.495	99.105	87.489
snowflake-arctic-embed-l-v2,0	63.73	43.632	50.25	77.25	40.04	96.383	74.843
gte-multilingual-base	62.12	52.358	42.25	55.5	43.033	95.499	84.078
bge-m3	58.73	45.761	40.25	59	35.543	94.089	77.726

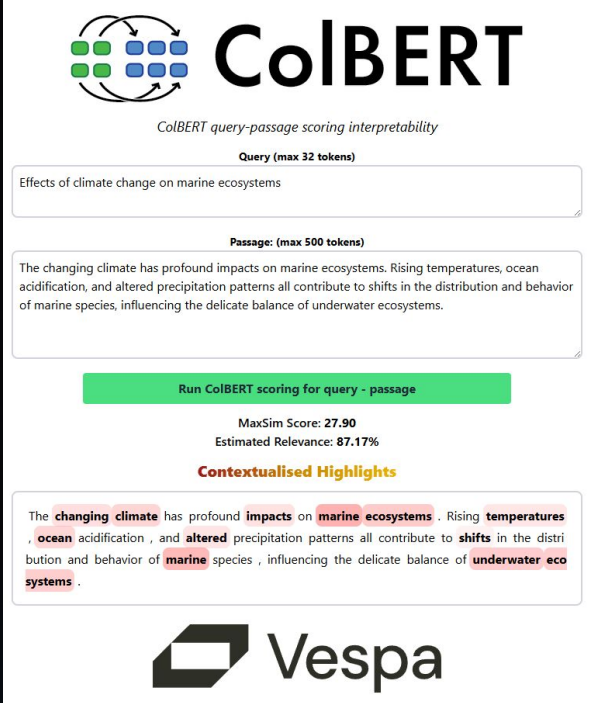
A simple, yet effective, difference

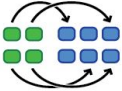
- GTE-ModernColBERT is SOTA on the Long Embed by a large margin while being trained on document length of 300
 - Performance improves even from 8k -> 16k while even the base model has been trained at 8k max
- Reason-ModernColBERT outperforms all 7B models (45 times bigger) on BRIGHT, competitive with ReasonIR-8B trained on the same data

Model / Metric	Biology	Earth	Economics	Psychology	Robotics	Stackoverflow	Sustainable	Leetcode	Pony	AoPS	Theorem - Q	Theorem - T	Mean StackExchange	Mean coding	Mean theorem	Full mean
BM25	18.9	27.2	14.9	12.5	13.6	18.4	15	24.4	7.9	6.2	10.4	4.9	17.21	16.15	7.17	14.53
< 1B OS																
BGE	11.7	24.6	16.6	17.5	11.7	10.8	13.3	26.7	5.7	6	13	6.9	15.17	16.2	8.63	13.71
> 1B OS																
E5	18.6	26	15.5	15.8	16.3	11.2	18.1	28.7	4.9	7.1	26.1	26.8	17.36	16.8	20	17.93
Qwen	30.6	36.4	17.8	24.6	13.2	22.2	14.8	25.5	9.9	14.4	27.8	32.9	22.8	17.7	25.03	22.51
Proprietary																
Cohere	18.7	28.4	20.4	21.6	16.3	18.3	17.6	26.8	1.9	6.3	15.7	7.2	20.19	14.35	9.73	16.6
OpenAI	23.3	26.7	19.5	27.6	12.8	14.3	20.5	23.6	2.4	8.5	23.5	11.7	20.67	13	14.57	17.87
Voyage	23.1	25.4	19.9	24.9	10.8	16.8	15.4	30.6	1.5	7.5	27.4	11.6	19.47	16.05	15.5	17.91
Google	22.7	34.8	19.6	27.8	15.7	20.1	17.1	29.6	3.6	9.3	23.8	15.9	22.54	16.6	16.33	20
ReasonIR data																
ReasonIR-8B	26.2	31.4	23.3	30	18	23.9	20.5	35	10.5	14.7	31.9	27.2	24.76	22.75	24.6	24.38
Reason-ModernColBERT (150M)	33.25	41.02	24.93	30.73	21.12	20.62	20.31	31.07	8.51	9.17	19.51	11.24	27.43	19.79	15.38	22.62

Interpretability: a nice little bonus

- Granular matching: understand why the document matched
- Identify sub-chunk of the document that matches



 **ColBERT**

ColBERT query-passage scoring interpretability

Query (max 32 tokens)

Effects of climate change on marine ecosystems

Passage: (max 500 tokens)

The changing climate has profound impacts on marine ecosystems. Rising temperatures, ocean acidification, and altered precipitation patterns all contribute to shifts in the distribution and behavior of marine species, influencing the delicate balance of underwater ecosystems.

Run ColBERT scoring for query - passage

MaxSim Score: 27.90
Estimated Relevance: 87.17%

Contextualised Highlights

The **changing climate** has profound **impacts** on **marine ecosystems**. Rising **temperatures**, **ocean** acidification, and **altered** precipitation patterns all contribute to **shifts** in the distribution and behavior of **marine** species, influencing the delicate balance of **underwater ecosystems**.

 **Vespa**

Why are people **still using** dense then?

- **Storing cost**: storing n tokens instead of one
 - Adapted indexes, quantization, footprint reduction techniques
- **Vectordb support**: not originally supported by vectordb providers
 - Most of them now support it (Vespa, Weaviate, QDrant, LanceDB, Milvus, Elastic)
- **Lack of accessible tools** to train, use and experiment with these models
 - Sentence transformers is a big reason why there is are many dense models!



What are the future avenues?

- Reduce the cost of multi-vector models, find the optimal trade-off

	Index Size
16-bit Dense Vector	345 MB
2-bit PLAID Index	760 MB
<i>pool 2</i>	388 MB
<i>pool 3</i>	260 MB
<i>pool 4</i>	195 MB
<i>pool 6</i>	131 MB

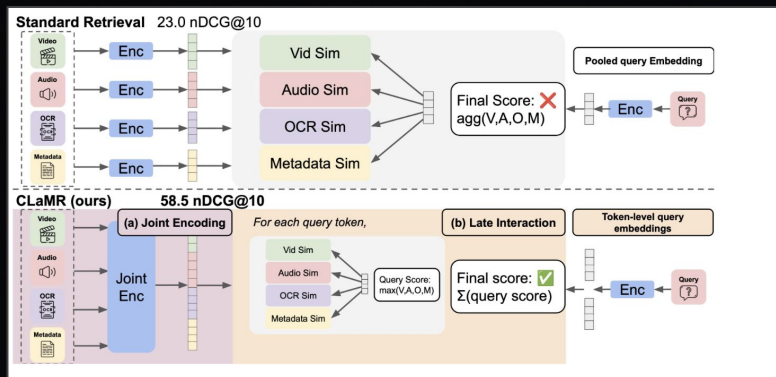
Table 3: An overview of index size (in Megabytes) between single-vector representation and PLAID ColBERT indexes.

	cfever	dbpedia	fever	fiqa	hotpot	nf	nq	scidocs	scifact	covid	touche	Avg.
nDCG@10												
ColBERTv2	0.163	0.434	0.751	0.338	0.679	0.329	0.554	0.154	0.638	0.705	0.261	0.455
Hierarch. Pooling ₂	0.170	0.438	0.744	0.331	0.657	0.327	0.553	0.153	0.629	0.719	0.294	0.456
Hierarch. Pooling ₃	0.168	0.414	0.740	0.311	0.631	0.310	0.526	0.152	0.614	0.730	0.321	0.447
Hierarch. Pooling ₄	0.171	0.407	0.738	0.285	0.604	0.310	0.503	0.148	0.578	0.734	0.333	0.437
Hierarch. Pooling ₆	0.171	0.374	0.731	0.258	0.544	0.293	0.468	0.144	0.574	0.707	0.348	0.419

What are the future avenues?

- Reduce the cost of multi-vector models, find the optimal trade-off
- Apply it to other modalities

This dual approach provides greater effectiveness with multi-vector representations, particularly in visually rich document retrieval, while maintaining efficiency for standard similarity tasks. The consistent 7-10% performance advantage of multi-vector over single-vector mode across visual tasks suggests that late interaction provides fundamentally better semantic matching for multimodal content.



What are the future avenues?

- Reduce the cost of multi-vector models, find the optimal trade-off
- Apply it to other modalities
- Find better similarity functions

Efficient Document Ranking with Learnable Late Interactions

Ziwei Ji Himanshu Jain Andreas Veit Sashank J. Reddi
Sadeep Jayasumana Ankit Singh Rawat Aditya Krishna Menon Felix Yu
Sanjiv Kumar

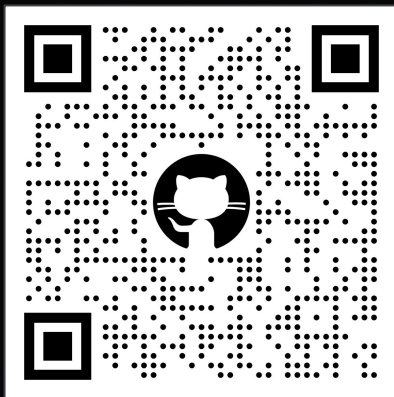
Google

{ziwei ji, himj, aveit, sashank, sadeep, ankitsrawat,
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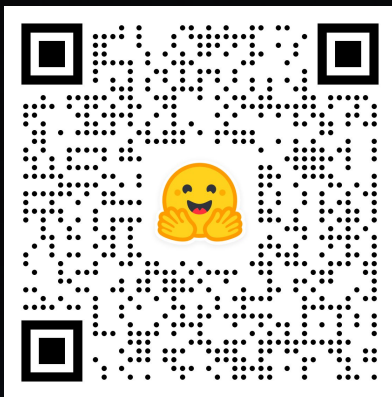
Conclusion

- Late interaction models overcome the **intrinsic limitations** of single vector search
- They are well-suited to your **modern real-world use cases** (out-of-domain generalization, long context, reasoning-intensive)
 - And probably a lot more to be discovered with new evaluations!
- They start to have a well supported ecosystem (e.g., vector providers)
- **PyLate** allows to easily train, experiment and evaluate those models with an experience very similar to sentence transformers
 - Go try existing models on your data and train your own specialized models!

Get started with **PyLate**



Try out **GTE-ModernColBERT**



Try out **Reason-ModernColBERT**

